ARTIFICIAL NEURAL NETWORKS

Loss Functions and Evaluation Metrics for Customer Churn Prediction

Project Domain: Binary Classification

Dataset: Telco Customer Churn

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Course
Artificial Neural Networks

PROJECT CONTEXT

Project Overview

Project Problem

Predict whether a telecom customer will leave the service (churn) based on historical data and behavioral patterns.

Goal of the Model

Identify customers likely to churn so companies can take preventive actions, implement retention strategies, and reduce revenue loss.

Machine Learning Task

Binary Classification – Two possible outcomes:

- Churn = Yes
- Churn = No

Example Features

- Tenure
- Monthly Charges
- Contract Type
- Internet Service
- Payment Method

FUNDAMENTAL CONCEPTS

Loss Functions vs Evaluation Metrics

Understanding the distinction between training optimization and performance assessment

Loss Functions

When Used

Applied **during model training** to guide the optimization process and update model parameters iteratively.

Purpose

- Measure the difference between predicted values and true values
- The training algorithm **minimizes the loss** to improve model parameters

> Loss functions guide model optimization throughout the training phase

Evaluation Metrics

When Used

Applied **after training** to assess how well the model performs on unseen validation and test datasets.

Purpose

- Evaluate how well the model performs on unseen data
- **Compare different models** and select the best performer

> Evaluation metrics measure prediction quality on held-out data

LOSS FUNCTION

Binary Cross Entropy (BCE)

The most common loss function for binary classification problems

Formula

$$L = -[\, y \cdot \log(p) + (1-y) \cdot \log(1-p) \,]$$

y = true label (0 or 1)

p = predicted probability

How It Works

If the model predicts the **wrong class with high confidence**, the loss increases significantly. This penalizes confident incorrect predictions more heavily than uncertain ones, encouraging the model to be both accurate and well-calibrated in its probability estimates.

Advantages

- ✓ **Probabilistic interpretation** – outputs can be interpreted as probabilities
- ✓ **Differentiable** – enables gradient-based optimization
- ✓ **Works well with neural networks** – stable gradients

Limitations

- ✕ **Sensitive to class imbalance** – dominated by majority class
- ✕ Can lead to **biased models** if one class dominates

The paper identifies BCE as a standard loss for binary classification tasks

ALTERNATIVE LOSS FUNCTIONS

Weighted Cross Entropy & Hinge Loss

Specialized loss functions for specific scenarios

Weighted Cross Entropy

An extension of Cross Entropy designed for **imbalanced datasets** where one class significantly outnumbers the other.

$$L = -[\, w_1 \cdot y \cdot \log(p) + w_0 \cdot (1-y) \cdot \log(1-p) \,]$$

where w_1 and w_0 are class weights

Why Useful

In churn datasets: **Non-churn customers >> Churn customers**. Without weighting, the model may ignore the minority class.

Advantages

- Improves minority detection
- Helps imbalanced data

Limitations

- Requires weight tuning
- Can destabilize training

Hinge Loss

Commonly used in **Support Vector Machines (SVM)** to create maximum margin classifiers.

$$L = \max(\, 0, 1 - y \cdot f(x) \,)$$

where y = true label, $f(x)$ = predicted score

Key Idea

Encourages the model to create a **margin between classes**, penalizing predictions that are too close to the decision boundary.

Advantages

- Strong decision boundaries
- Effective margin-based

Limitations

- Not probabilistic
- Less common in neural nets

EVALUATION METRIC

Accuracy

The proportion of correct predictions

Formula

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

- **TP** = True Positive
- **TN** = True Negative
- **FP** = False Positive
- **FN** = False Negative

Advantages

- **Easy to understand** – intuitive interpretation
- **Simple evaluation metric** – quick assessment

Critical Limitation

Misleading for Imbalanced Data

Accuracy can give a false sense of model performance when classes are imbalanced.

Example Scenario

If **90% of customers do not churn**, predicting "No Churn" for everyone gives **90% accuracy** – a useless model appears excellent!

Confusion Matrix Visualization

True Positive TP	False Positive FP
False Negative FN	True Negative TN

EVALUATION METRICS

Precision & Recall

Complementary metrics for evaluating positive class predictions

Precision

Measures **how many predicted positive cases are actually correct** – the reliability of positive predictions.

$$\text{Precision} = TP / (TP + FP)$$

Interpretation

"Of all customers predicted to churn, how many actually churn?"

+ Advantage

Useful when **false positives are costly**

– Limitation

Does not consider false negatives

Recall

Measures **how many actual positives are detected** – the completeness of positive predictions.

$$\text{Recall} = TP / (TP + FN)$$

Interpretation

"Of all customers who churned, how many were correctly identified?"

+ Advantage

Important when **missing positives is costly**

– Limitation

High recall can increase false positives

COMPREHENSIVE EVALUATION METRICS

F1 Score & ROC-AUC

Balanced and threshold-independent performance measures

F1 Score

The **harmonic mean of precision and recall**, providing a balanced measure that considers both false positives and false negatives.

$$F1 = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Advantages

- Balances precision and recall
- Good for imbalanced datasets

Limitations

- Harder to interpret than accuracy
- Does not consider true negatives

ROC Curve & AUC

ROC Curve

Plots **True Positive Rate vs False Positive Rate** at various threshold settings

AUC (Area Under Curve)

Measures **overall model discrimination ability**

Perfect Model	Random Guessing
AUC = 1.0	AUC = 0.5

Advantages

- Evaluates discrimination ability
- Works well for imbalanced data

The paper lists AUC-ROC as a key classification evaluation metric

COMPREHENSIVE COMPARISON

Comparative Analysis

Side-by-side comparison of loss functions and evaluation metrics

Comparison of Loss Functions

Loss Function	Use Case	Advantages	Limitations
Binary Cross Entropy	Binary classification	Probabilistic interpretation	Sensitive to imbalance
Weighted Cross Entropy	Imbalanced datasets	Improves minority detection	Requires weight tuning
Hinge Loss	Margin-based models	Strong decision boundaries	Not probabilistic

Comparison of Evaluation Metrics

Metric	Advantage	Limitation
Accuracy	Easy to understand	Misleading for imbalanced data
Precision	Reduces false positives	Ignores false negatives
Recall	Detects positives	Can increase false positives
F1 Score	Balanced evaluation	Harder to interpret
ROC-AUC	Measures class separation	Less intuitive

PROJECT IMPLEMENTATION

Metrics for Our Churn Prediction Project

Comprehensive evaluation framework for model assessment

Loss Function

For model training

Binary Cross Entropy

Standard loss for binary classification

Evaluation Metrics

For model performance assessment

Accuracy

Overall correctness

Precision

Prediction reliability

Recall

Completeness

F1 Score

Balanced measure

ROC-AUC

Discrimination ability across all thresholds

Why This Selection?

This combination provides a **comprehensive evaluation of classification performance** across different aspects: overall correctness (accuracy), prediction reliability (precision), completeness (recall), balance (F1), and discrimination ability (ROC-AUC).

KEY TAKEAWAYS

Conclusion

Essential insights for building reliable churn prediction systems

1 Loss Functions Guide Training

Loss functions are used **during model training** to measure prediction error and guide parameter optimization through gradient descent.

2 Metrics Measure Performance

Evaluation metrics are used **after training** to assess model quality on unseen data and enable fair model comparison.

3 BCE is the Standard

Binary Cross Entropy is the most commonly used loss function for binary classification due to its probabilistic interpretation and differentiability.

4 Comprehensive Evaluation

Metrics like **Precision, Recall, F1 Score, and ROC-AUC** collectively provide a holistic view of model performance.

Building Reliable Systems

These methods are **essential for building reliable customer churn prediction systems** that enable proactive customer retention strategies and reduce business revenue loss.